IBM Developer
SKILLS NETWORK

# Winning Space Race with Data Science

**Vikas Parmar**

**June 14, 2026**

# Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

# Executive Summary

Summary of Methodologies

1. Data Collection: Accessed SpaceX launch data via API and web scraped records from Wikipedia.
2. Data Cleaning & Preparation:
    1. Cleaned and formatted the data.
    2. Stored data in Db2 database and performed SQL queries.
    3. Conducted exploratory data analysis.
3. Feature Engineering: Created new features and standardized the data.
4. Interactive Visualizations:
    1. Mapped launch sites and success rates using Folium.
    2. Built an interactive dashboard with Plotly Dash. Executive Summary
5. Model Building & Evaluation:
    1. Implemented SVM, Decision Trees, and K-Nearest Neighbors.
    2. Tuned hyperparameters with GridSearchCV.
    3. Evaluated models using test data accuracy. Summary of Results 1. Data Insights:

*Vikas Parmar*

# Introduction

**Project Background and Context:** In this capstone project, we aim to predict the successful landing of the Falcon 9 first stage. SpaceX advertises rocket launches at a significantly lower cost compared to other providers, largely due to their ability to reuse the first stage of the rocket. By accurately predicting landing success, we can estimate launch costs and provide valuable insights for companies bidding against SpaceX.

- SpaceX advertises Falcon 9 launches at about $62M versus $165M+ from other providers. Reusing the first stage drives much of that savings.
- **Problem:** For a given launch (payload, orbit, site, booster version), will the Falcon 9 first stage land successfully?
- **Audience:** A competing aerospace startup estimating launch bid costs against SpaceX.

Section 1

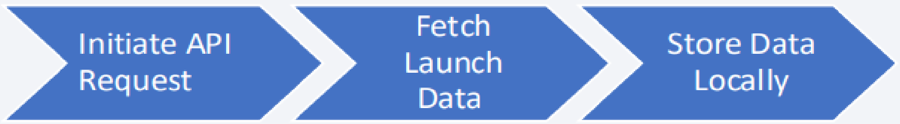
# Methodology

# Methodology

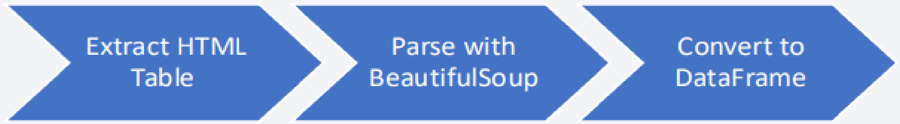
- **Executive Summary:** This project employs a comprehensive approach to predict the successful landing of the Falcon 9 first stage, incorporating data collection, processing, exploratory analysis, interactive visualizations, and predictive modelling.
- **Data Collection Methodology:** Data was sourced from the SpaceX API, which provided detailed records of Falcon 9 launches, including launch dates, sites, payloads, and outcomes.
- **Data wrangling:** Missing values, outcome mapping, Class label
- **EDA:** SQL queries + Matplotlib/Seaborn charts
- **Interactive analytics:** Folium maps + Plotly Dash dashboard
- **Predictive analysis:** StandardScaler, GridSearchCV, 4 classifiers

# Data Collection

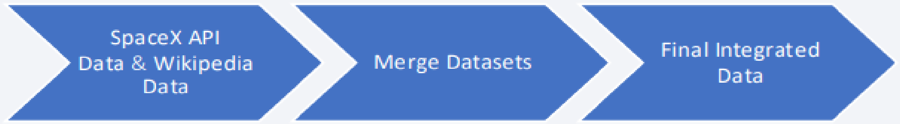
- Step 1: SpaceX API Request




- Step 2: Web Scraping Wikipedia



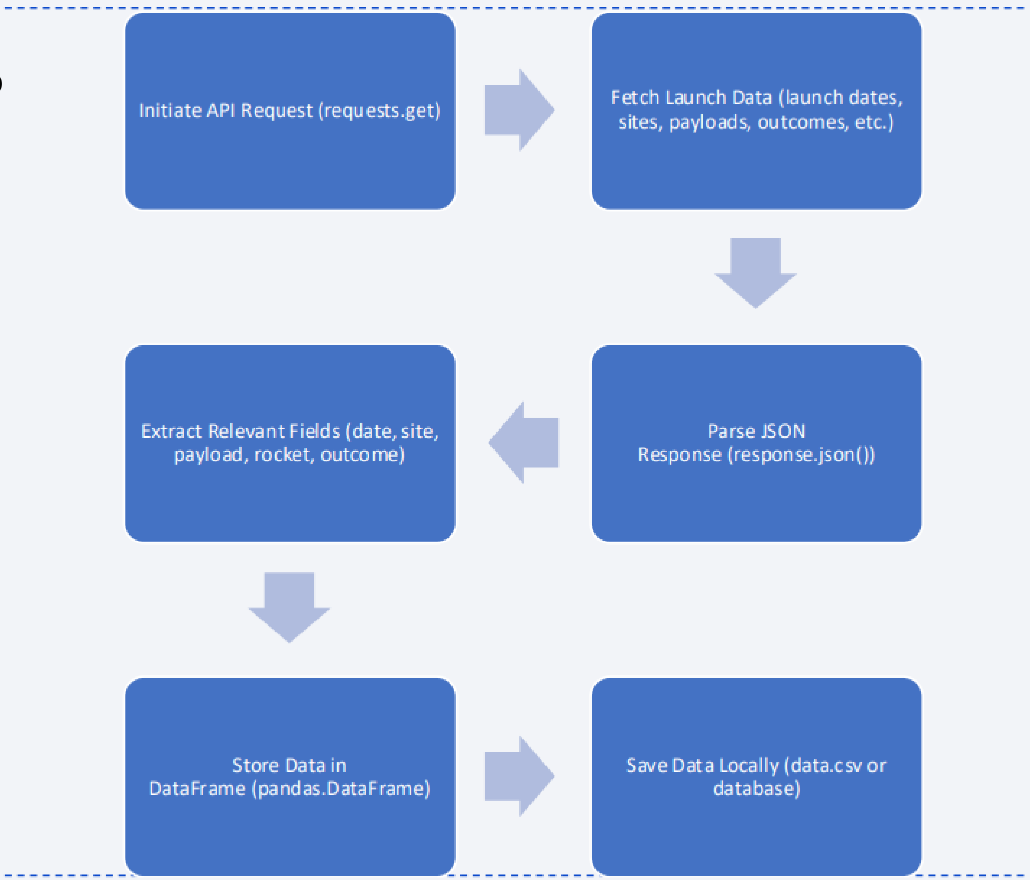

- Step 3: Data Integration



# Data Collection – SpaceX API

- Step 1: Initiate API Request
  - Use Python's `requests` library to connect to the SpaceX API. o
  - Endpoint: `https://api.spacexdata.com/v4/launches`
- Step 2: Parse API Response
  - Convert API response from JSON to a Python dictionary.
  - Extract relevant fields: launch date, launch site, payload mass, rocket type, outcome.
- Step 3: Store Data Locally
  - Save extracted data into a pandas DataFrame.
  - Store the DataFrame locally for further processing.

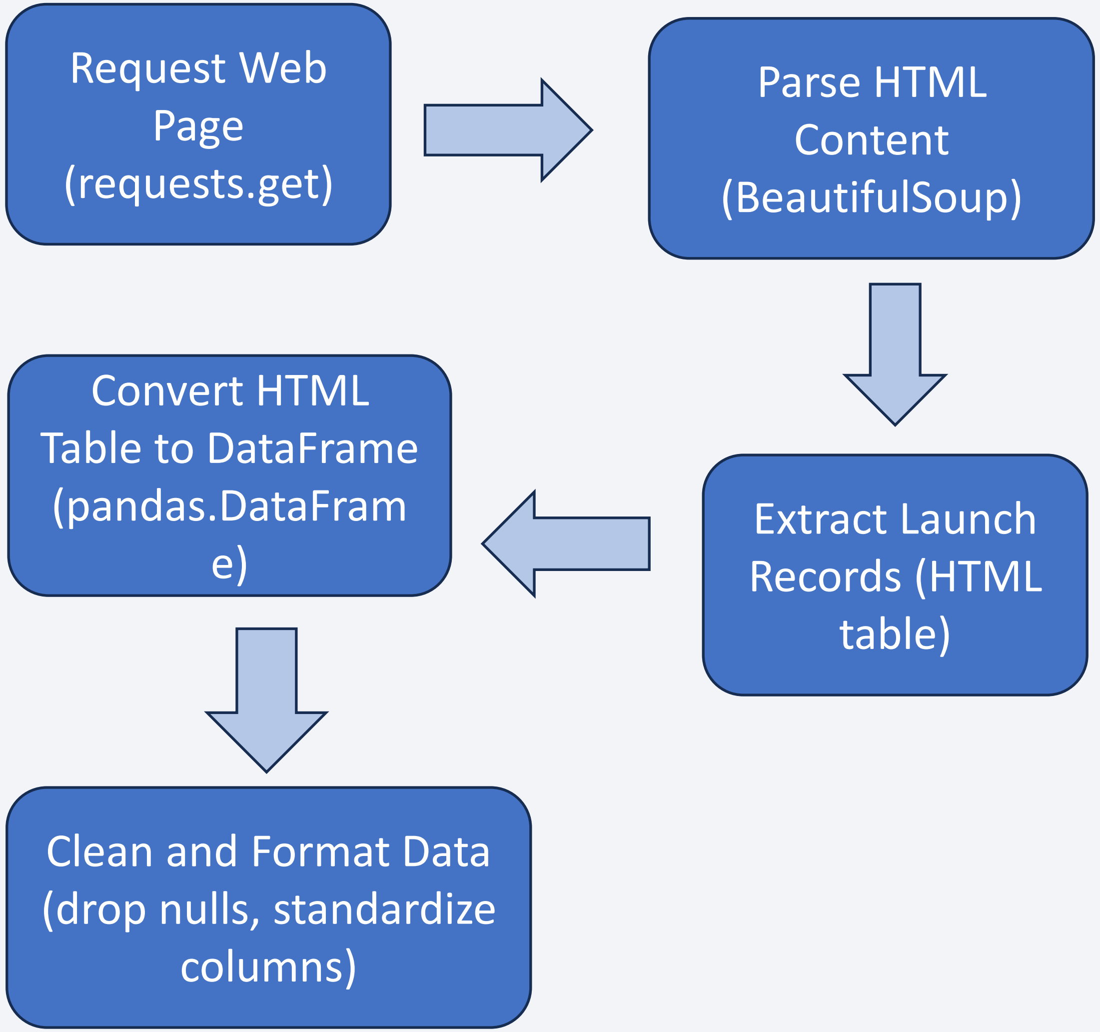
**GitHub**: https://github.com/vikas-parmar/applied-data-science-capstone-ibm/blob/main/Data%20Collection%20API.ipynb



# Data Collection - Scraping

1. **Initiate Web Scraping**
    1. Use Python's `requests` library to fetch the HTML content of the Wikipedia page.
    2. Target URL: `https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches`
2. **Parse HTML Content**
    1. Use `BeautifulSoup` to parse the HTML content.
    2. Extract the HTML table containing Falcon 9 launch records.
3. **Convert to DataFrame**
    1. Convert the extracted HTML table into a pandas DataFrame.
    2. Clean and format the DataFrame, ensuring data consistency

**GitHub**: https://github.com/vikas-parmar/applied-data-science-capstone-ibm/blob/main/Data%20Collection%20with%20Web%20Scraping.ipynb




*Vikas Parmar*

# Data Wrangling

**Overview:** Data wrangling involves cleaning, transforming, and organizing raw data into a structured format suitable for analysis.

- Step 1: Data Cleaning
  - Identify and fill or remove missing values in the dataset.
  - Use appropriate imputation techniques or drop rows/columns with excessive missing data.
- Step 2: Data Transformation
  - Convert data types to appropriate formats (e.g., date-time, numerical).
  - Standardize text (e.g., lowercase, remove whitespace).

# Data Wrangling

  - Create new features from existing data (e.g., extract year from date).
  - Normalize/scale numerical features to ensure consistency. Data Wrangling 12
- Step 3: Data Integration
  - Merge datasets collected from different sources (API, web scraping) into a single cohesive dataset.
  - Ensure consistent column names and data formats across datasets.
- Step 4: Data Validation
  - Check for duplicate records and remove them.
  - Verify the accuracy and consistency of data entries.

GitHub: https://github.com/vikas-parmar/applied-data-science-capstone-ibm/blob/main/Data%20Wrangling.ipynb

# EDA with Data Visualization

- **Overview:** Exploratory Data Analysis (EDA) involves visually exploring and summarizing the main characteristics of a dataset. The goal is to understand the data's distribution, identify patterns, and uncover relationships between variables.

**Charts Plotted:**

1. **Histograms: -**

- **Purpose:** Used to visualize the distribution of numerical variables such as launch success rates, payload mass, and flight number.
- **Why:** Helps in understanding the spread and central tendency of the data, identifying outliers, and assessing data skewness.

# EDA with Data Visualization

**2. Bar Charts: -**

- **Purpose:** Used to compare categorical variables such as launch outcomes (success/failure) across different categories like launch sites or rocket types.
- **Why:** Provides a clear comparison of frequencies or proportions within categorical data, highlighting patterns or trends.

**3. Line Charts: -**

- **Purpose:** Used to track trends over time, such as the success rate of Falcon 9 launches across different years.
- **Why:** Reveals temporal patterns and helps in understanding performance trends or changes over specific periods.

# EDA with Data Visualization

**4. Scatter Plots: -**

**- Purpose:** Used to explore relationships between two numerical variables, such as payload mass vs. launch success.

- **Why:** Identifies correlations or dependencies between variables, visualizing how one variable changes concerning another.
- **5. Heatmaps:**
- **Purpose:** Used to visualize correlation matrices between multiple numerical variables.
- **Why:** Helps in identifying strong correlations (positive or negative) between variables, aiding feature selection or understanding multicollinearity.

# EDA with Data Visualization

**6. Box Plots: -**

**- Purpose:** Used to display the distribution of numerical data through their quartiles.

- **Why:** Visualizes the spread and skewness of data, highlighting outliers and comparing distributions across different categories.

GitHub: https://github.com/vikas-parmar/applied-data-science-capstone-ibm/blob/main/EDA%20with%20Data%20Visualization.ipynb

# EDA with SQL

**Aggregate Queries:**
• Calculated total number of launches.
• Counted successful and failed launches.
• Calculated success rates by launch site and rocket type.

**Join Queries:**
• Joined tables to link launch records with additional data (e.g., rocket details).
• Combined datasets for comprehensive analysis.

**Filtering Queries:**
• Filtered data to focus on specific launch outcomes (success/failure).
• Applied conditions to extract launches based on criteria like launch date or rocket configuration.

**Sorting Queries:**
• Sorted data to identify trends or outliers.
• Ordered launches by date or success rate for analysis.

**Subqueries:**
• Nested queries to calculate derived metrics (e.g., average payload mass per launch site).
• Subqueries used to perform detailed analysis within larger datasets.

GitHub: https://github.com/vikas-parmar/applied-data-science-capstone-ibm/blob/main/EDA.ipynb

# Build an Interactive Map with Folium

Folium map objects:
- Markers for all launch sites
- Color-coded success/failure launch markers
- Proximity lines to coastline, highway, railway, and city

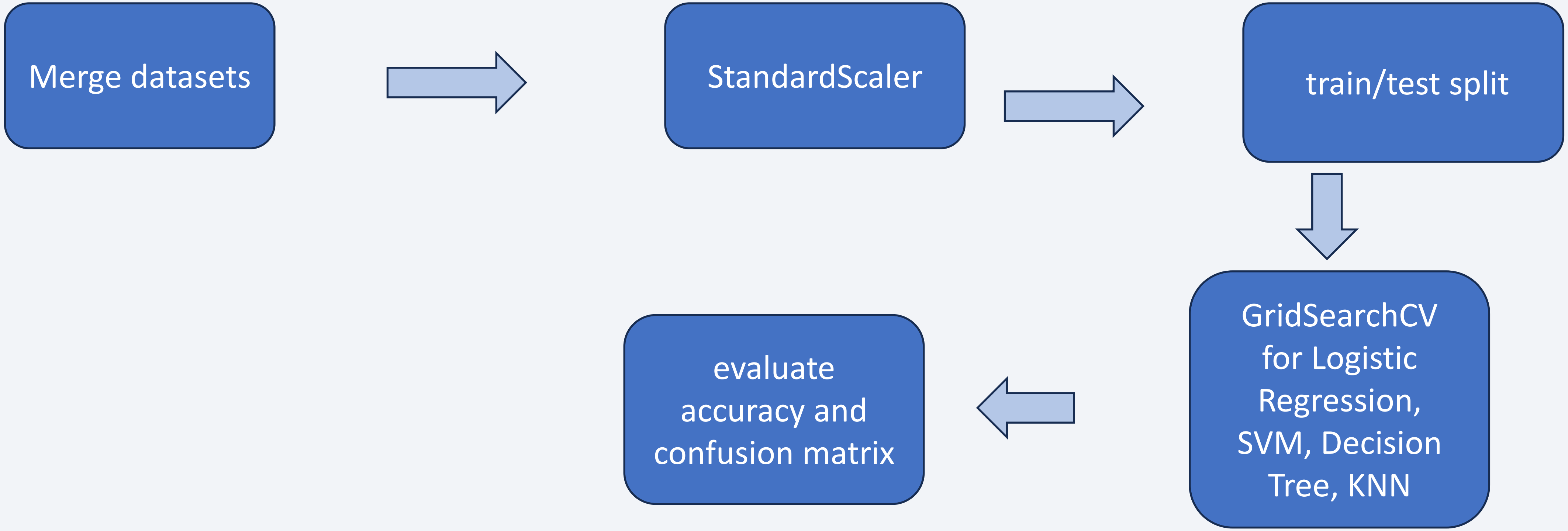
GitHub: https://github.com/vikas-parmar/applied-data-science-capstone-ibm/blob/main/Interactive%20Visual%20Analytics%20with%20Folium%20lab.ipynb

# Build a Dashboard with Plotly Dash

Plotly Dash dashboard:
- Launch site dropdown
- Success pie chart
- Payload range slider
- Payload vs launch outcome scatter plot

GitHub: https://github.com/vikas-parmar/applied-data-science-capstone-ibm/blob/main/spacex_dash_app.py

# Predictive Analysis (Classification)




GitHub: https://github.com/vikas-parmar/applied-data-science-capstone-ibm/blob/main/Machine%20Learning%20Prediction.ipynb

# Results

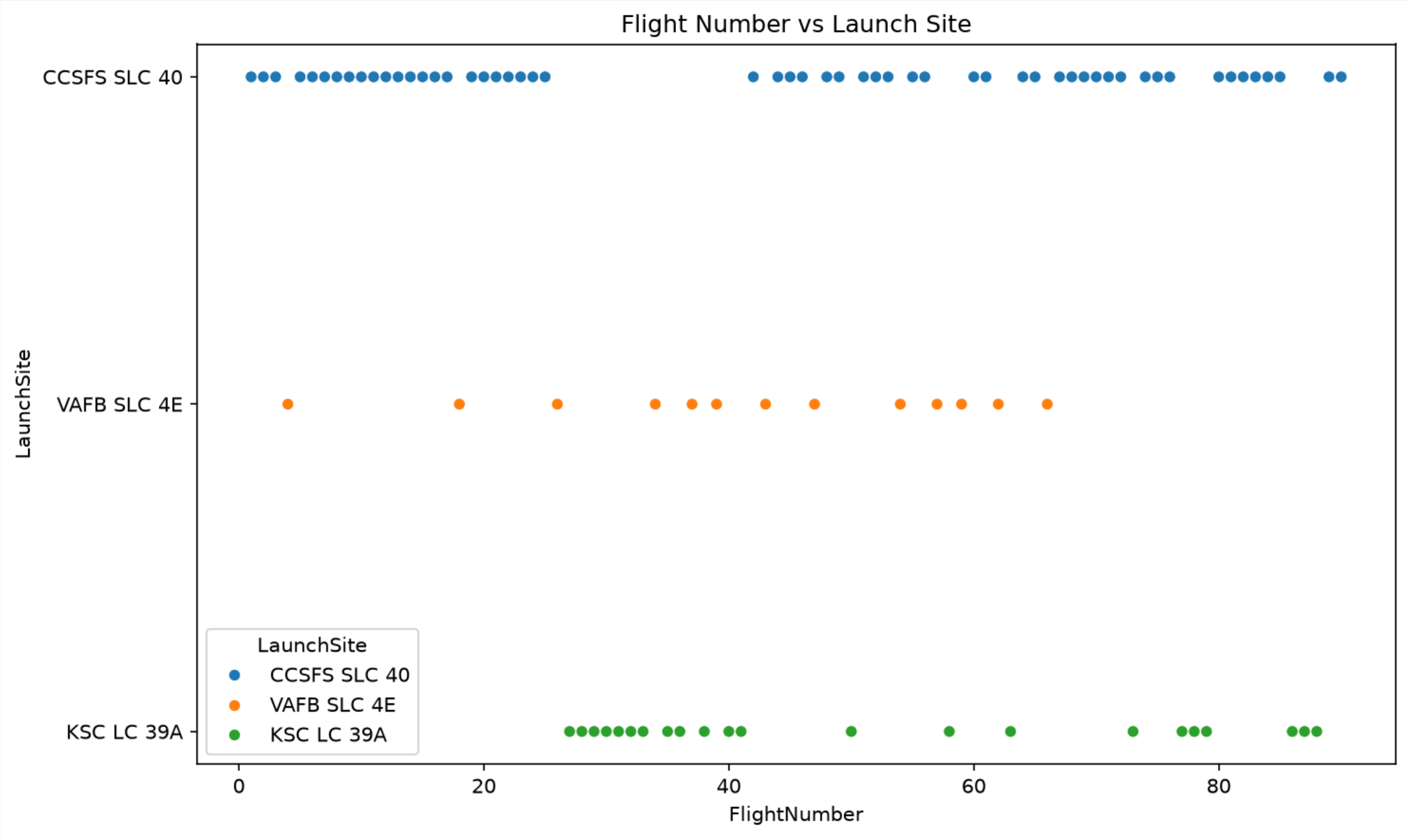
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

# Insights drawn from EDA

*Vikas Parmar*

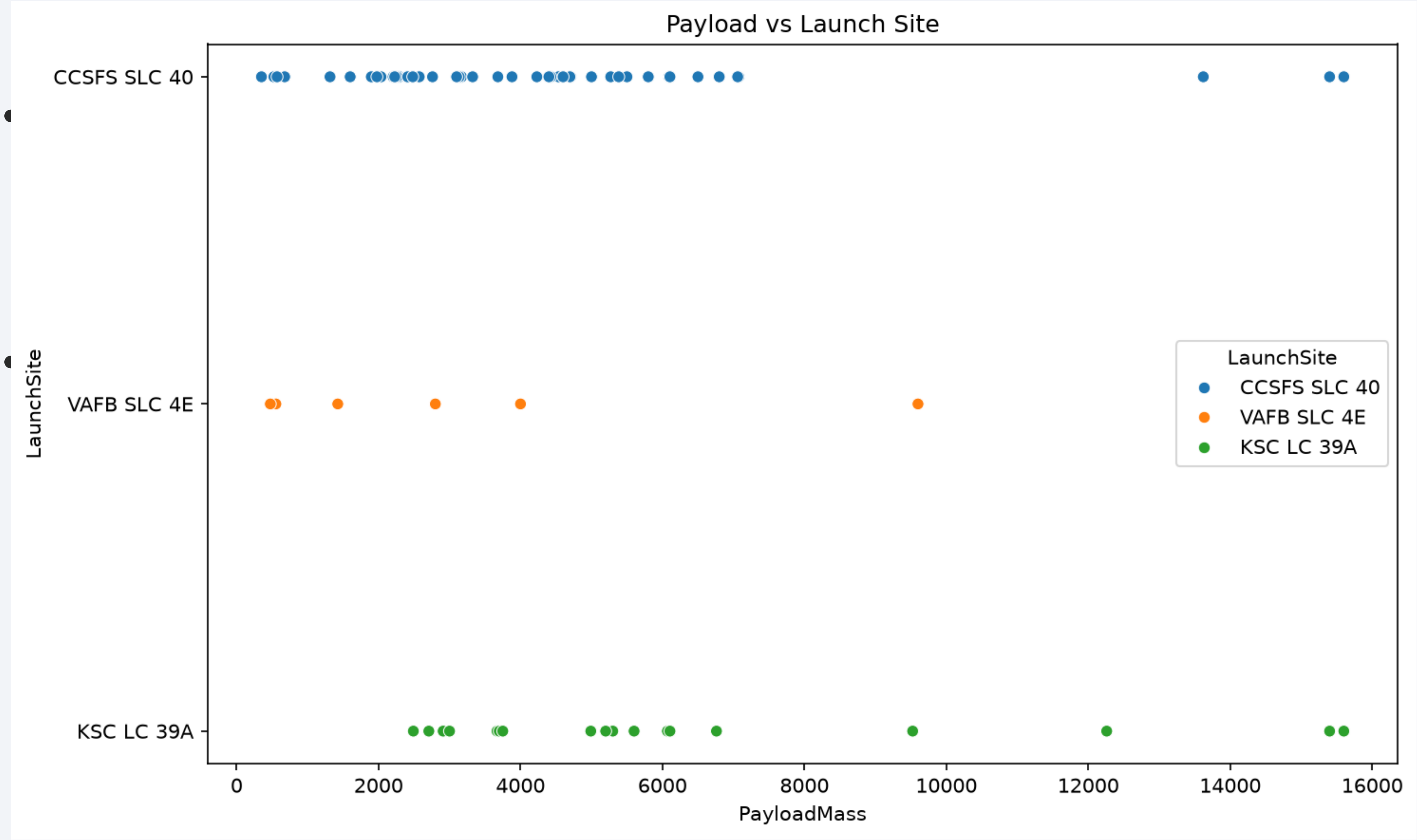
# Flight Number vs. Launch Site




Flight number increases over time at each launch site.

*Vikas Parmar*

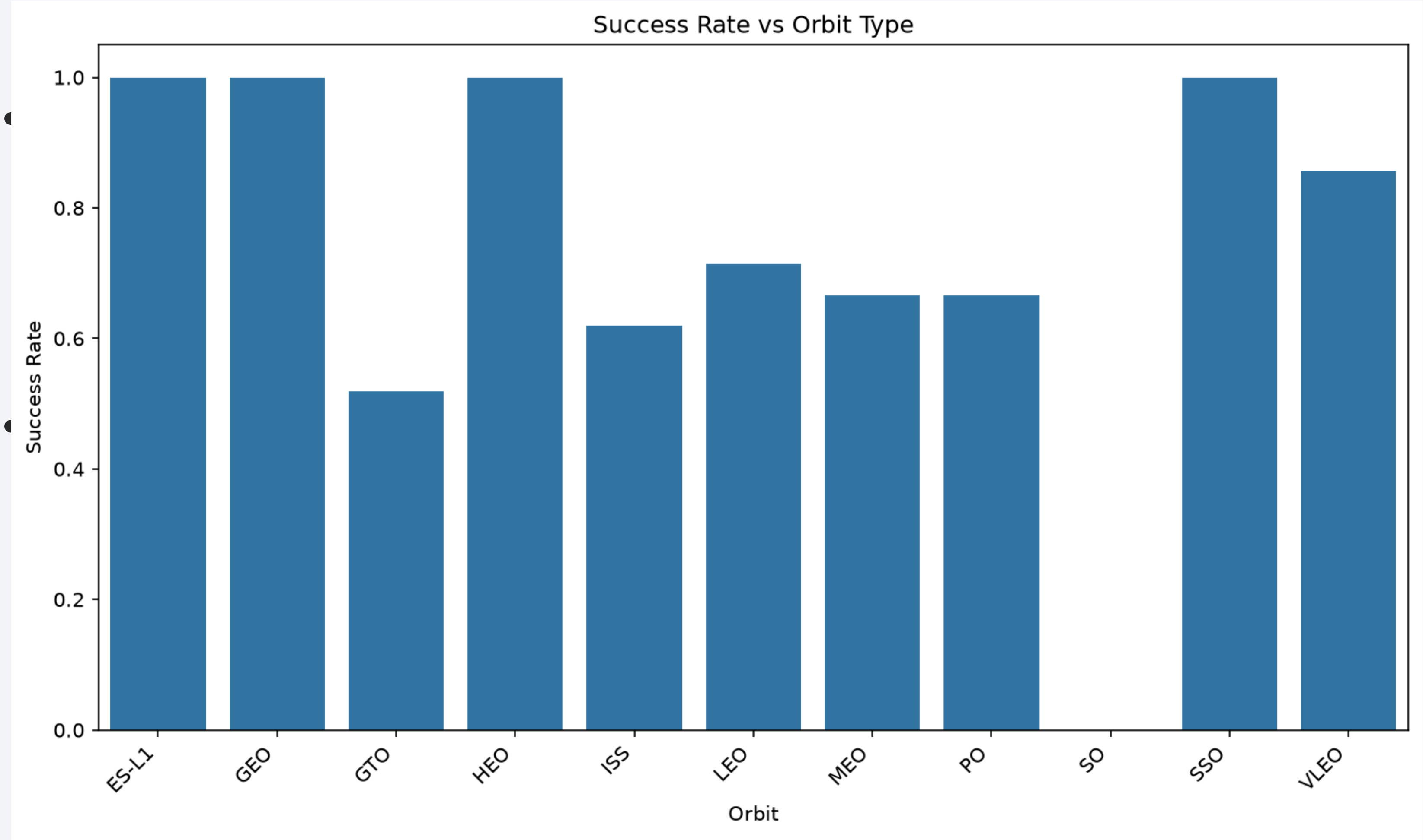
# Payload vs. Launch Site




Payload distribution varies significantly by launch site.

*Vikas Parmar*

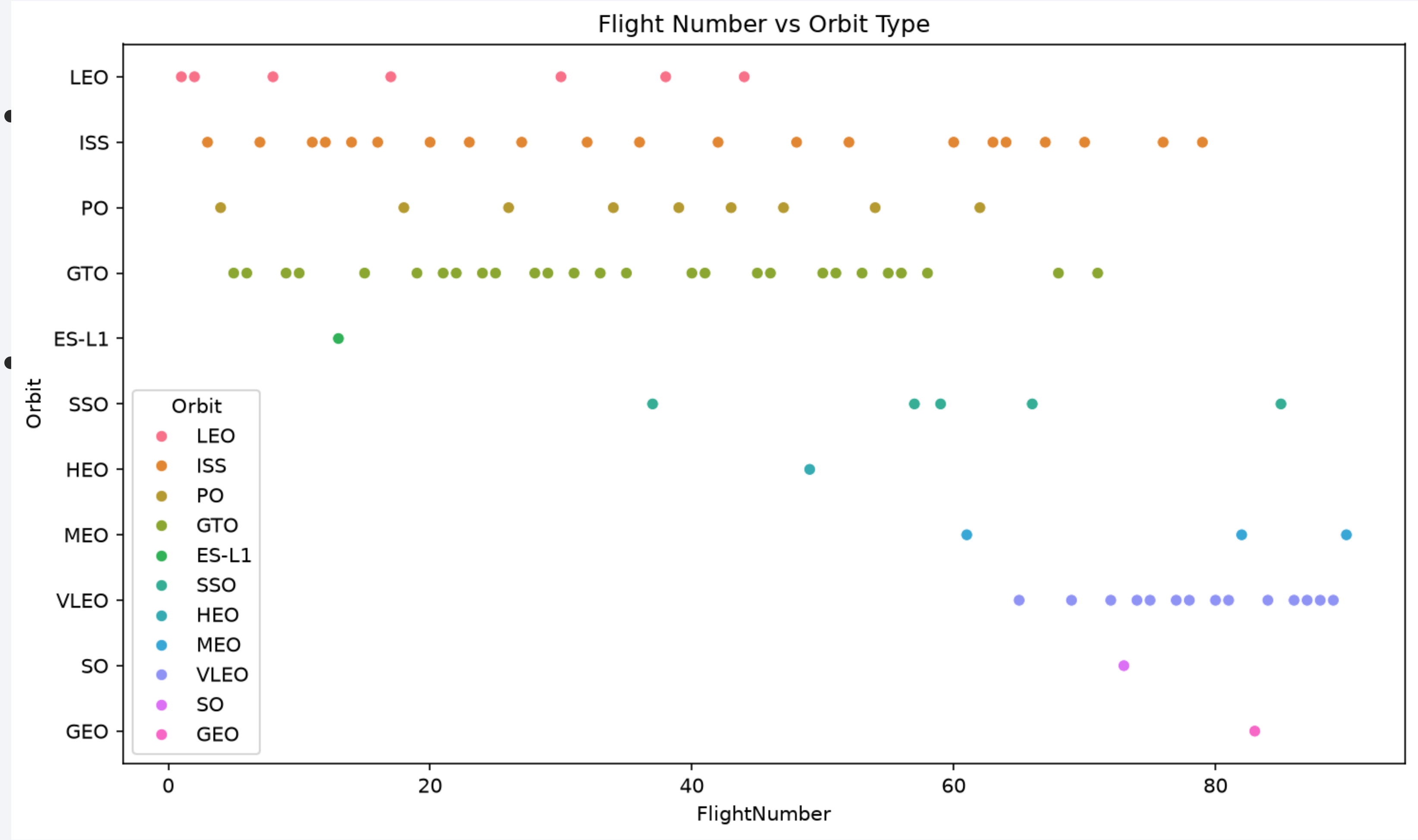
# Success Rate vs. Orbit Type




LEO and ISS orbits show higher success rates than GTO.

Vikas Parmar

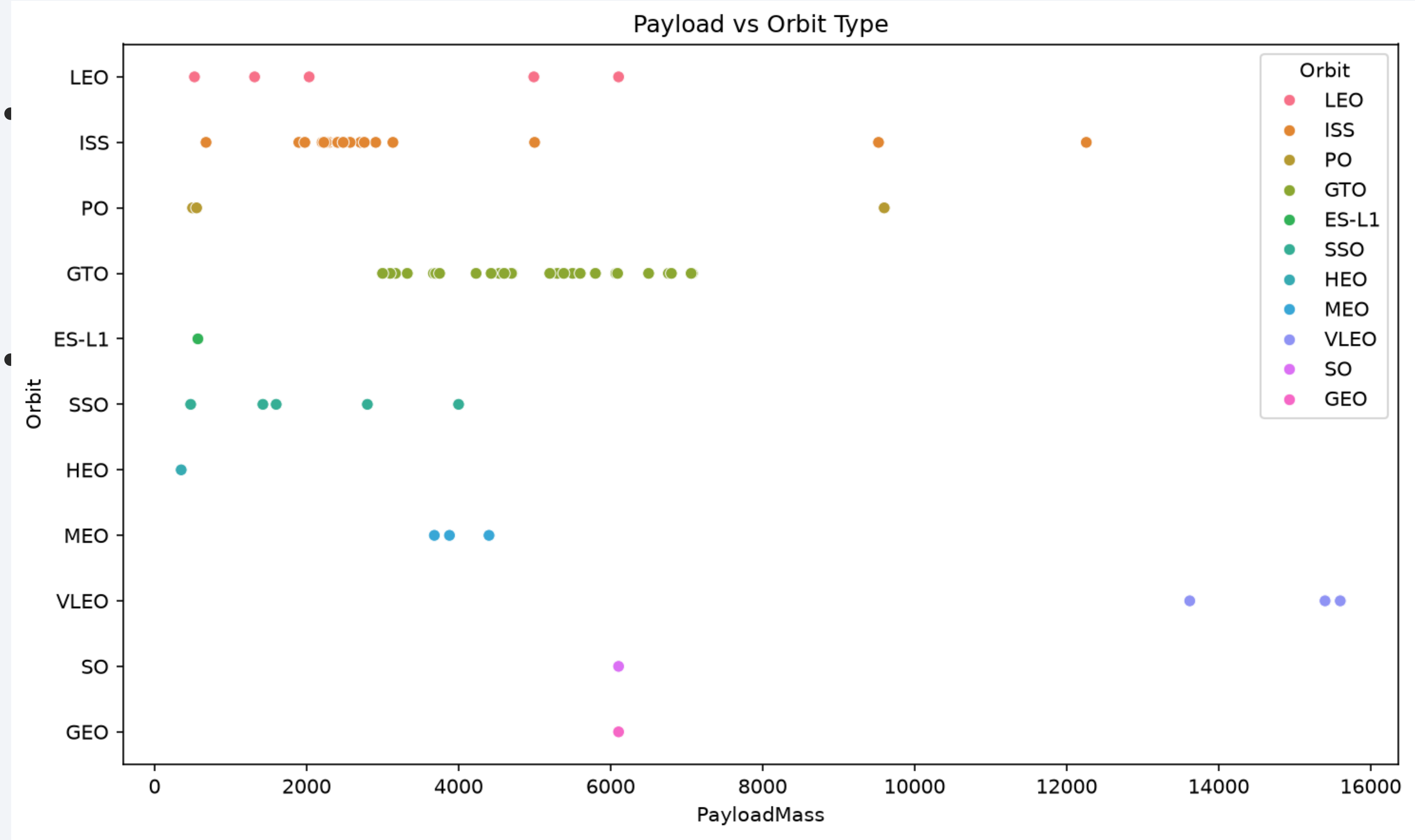
# Flight Number vs. Orbit Type




Later flights show more diverse orbit assignments.

*Vikas Parmar*

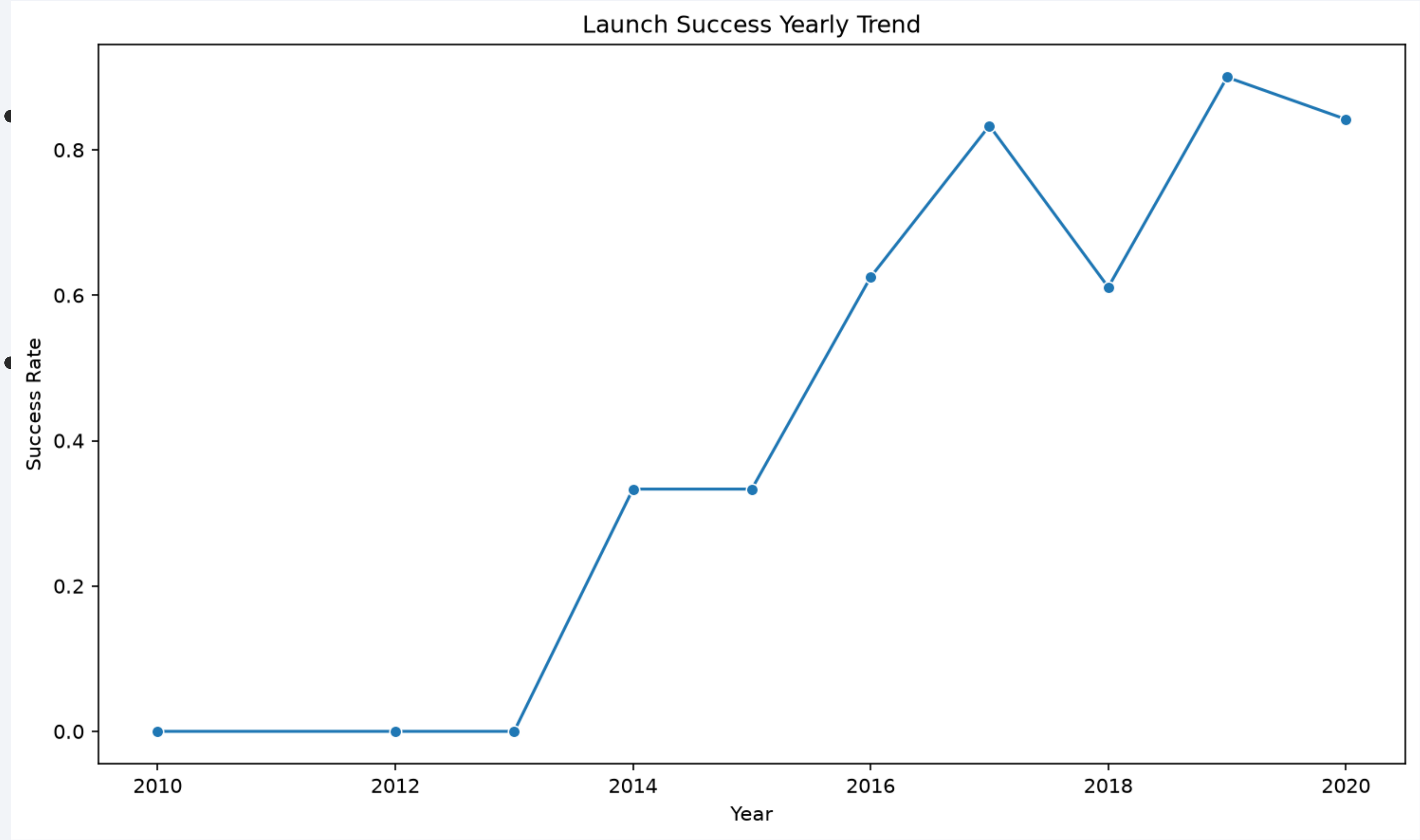
# Payload vs. Orbit Type




Heavier payloads appear more often in GTO missions.

*Vikas Parmar*

# Launch Success Yearly Trend




Landing success rate improved sharply after 2014.

# All Launch Site Names

Sql 01 Launch Sites

| LaunchSite |
| --- |
| CCAFS SLC 40 |
| VAFB SLC 4E |
| KSC LC 39A |

Four unique launch sites identified.

*Vikas Parmar*

# Launch Site Names Begin with 'CCA'

Sql 02 Cca Sites

| LaunchSite |
| --- |
| CCAFS SLC 40 |
| CCAFS SLC 40 |
| CCAFS SLC 40 |
| CCAFS SLC 40 |
| CCAFS SLC 40 |

CCAFS launch sites begin with CCA prefix.

*Vikas Parmar*

# Total Payload Mass

Sql 03 Nasa Payload

| total_payload_mass |
| --- |
| |

Total NASA CRS payload mass calculated.

Vikas Parmar

# Average Payload Mass by F9 v1.1

Sql 04 Avg F9V11

| avg_payload_mass |
|---|
| |

Average payload for F9 v1.1 boosters.

*Vikas Parmar*

# First Successful Ground Landing Date

Sql 05 First Ground Success

| first_successful_ground_landing |
|---|
| 2015-12-22 00:00:00 |

First successful RTLS landing date.

Vikas Parmar

# Successful Drone Ship Landing with Payload between 4000 and 6000

Sql 06 Drone Ship Payload

| BoosterVersion |
|---|
| Falcon 9 |
| Falcon 9 |
| Falcon 9 |
| Falcon 9 |
| Falcon 9 |
| Falcon 9 |

Boosters with drone ship landing and mid-range payload.

# Total Number of Successful and Failure Mission Outcomes

Sql 07 Mission Outcomes

| success_count | failure_count |
| --- | --- |
| 60 | 30 |

Successful vs failed mission counts.

# Boosters Carried Maximum Payload

Sql 08 Max Payload

| BoosterVersion | PayloadMass |
|---|---|
| Falcon 9 | 15600.0 |
| Falcon 9 | 15600.0 |
| Falcon 9 | 15600.0 |

Boosters carrying maximum payload mass.

# 2015 Launch Records

Sql 09 Failed 2015

| BoosterVersion | LaunchSite |
|---|---|
| Falcon 9 | CCAFS SLC 40 |
| Falcon 9 | CCAFS SLC 40 |

Failed drone ship landings in 2015.

*Vikas Parmar*

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Sql 10 Rank Outcomes

| Outcome | outcome_count |
|---|---|
| None None | 9 |
| True ASDS | 5 |
| False ASDS | 4 |
| True RTLS | 3 |
| True Ocean | 3 |
| None ASDS | 2 |
| False Ocean | 2 |

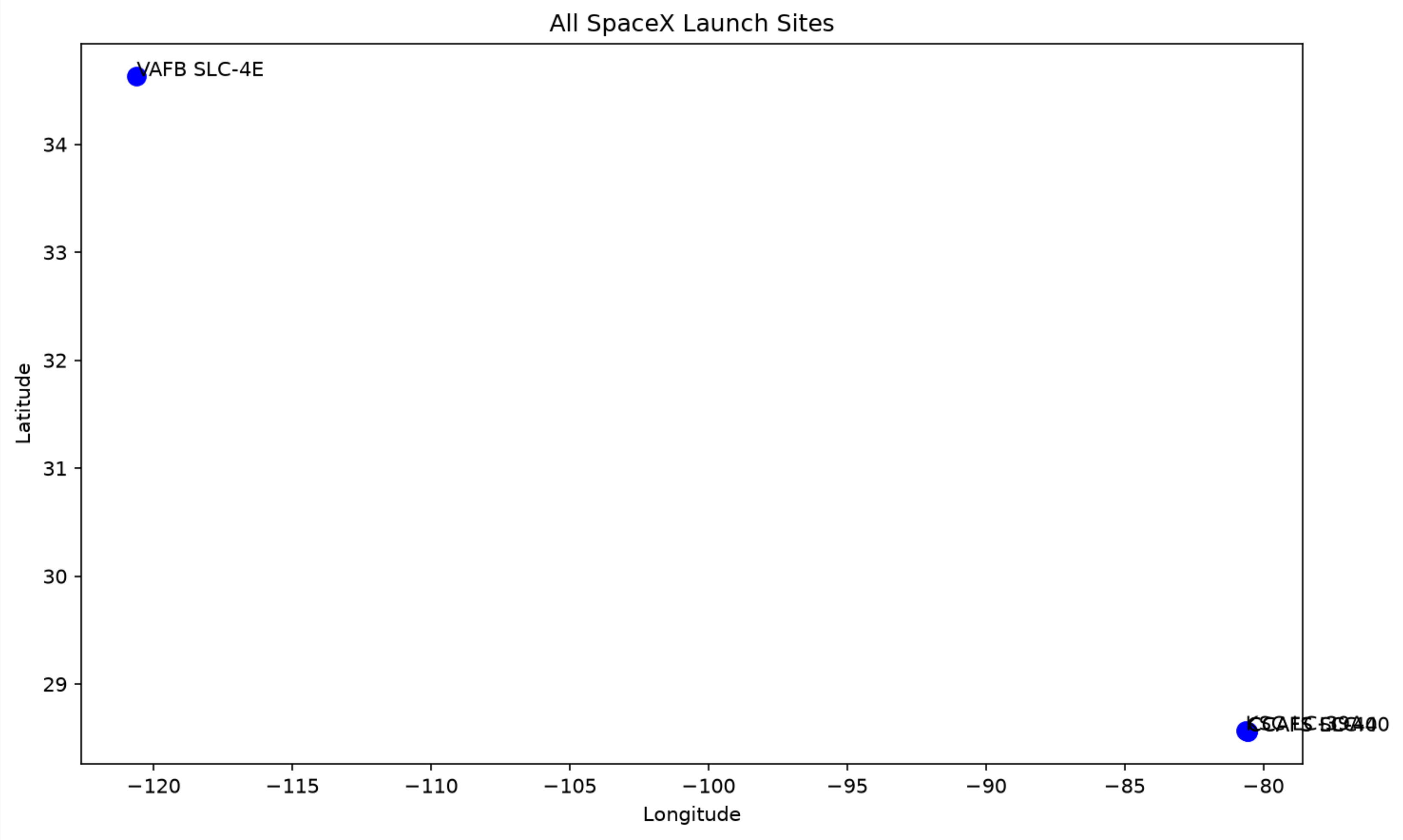
Ranked landing outcomes between 2010 and 2017.

Section 3

# Launch Sites Proximities Analysis

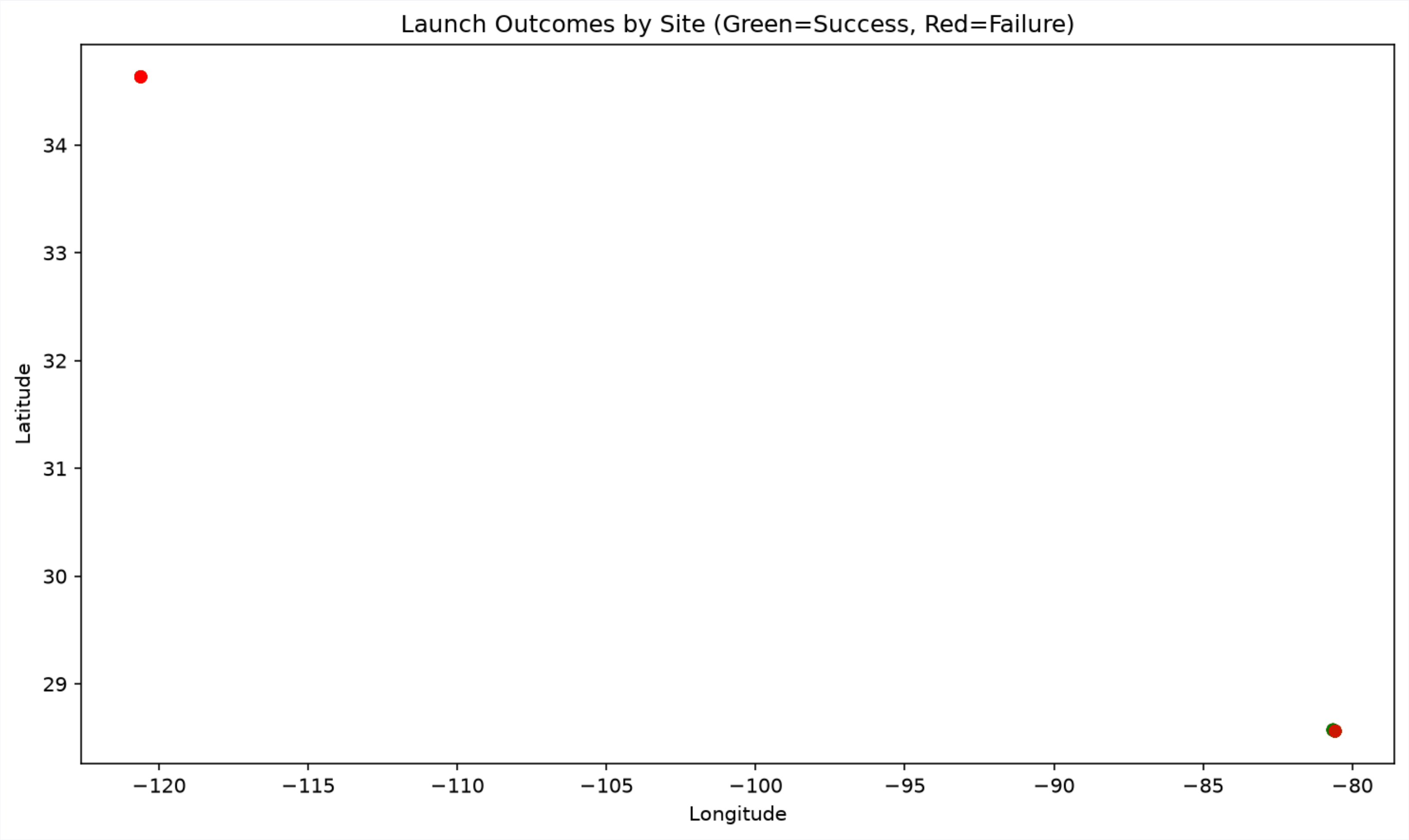
*Vikas Parmar*

# Folium Map Screenshot 1




All four SpaceX launch sites marked on the map.

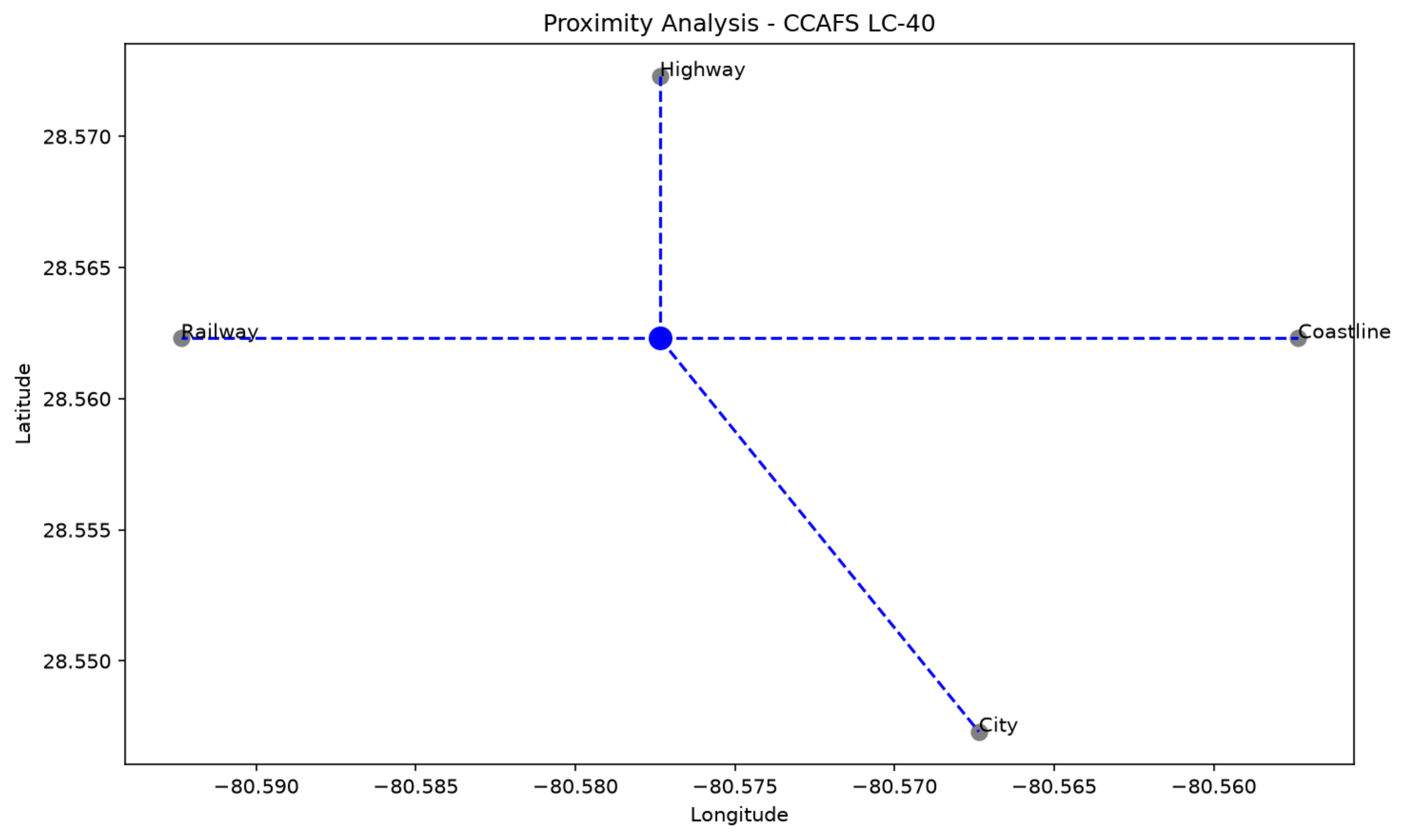
# Folium Map Screenshot 2




Green markers = successful landings; red = failures.

Vikas Parmar

# Folium Map Screenshot 3




Proximity analysis shows distance to key infrastructure.

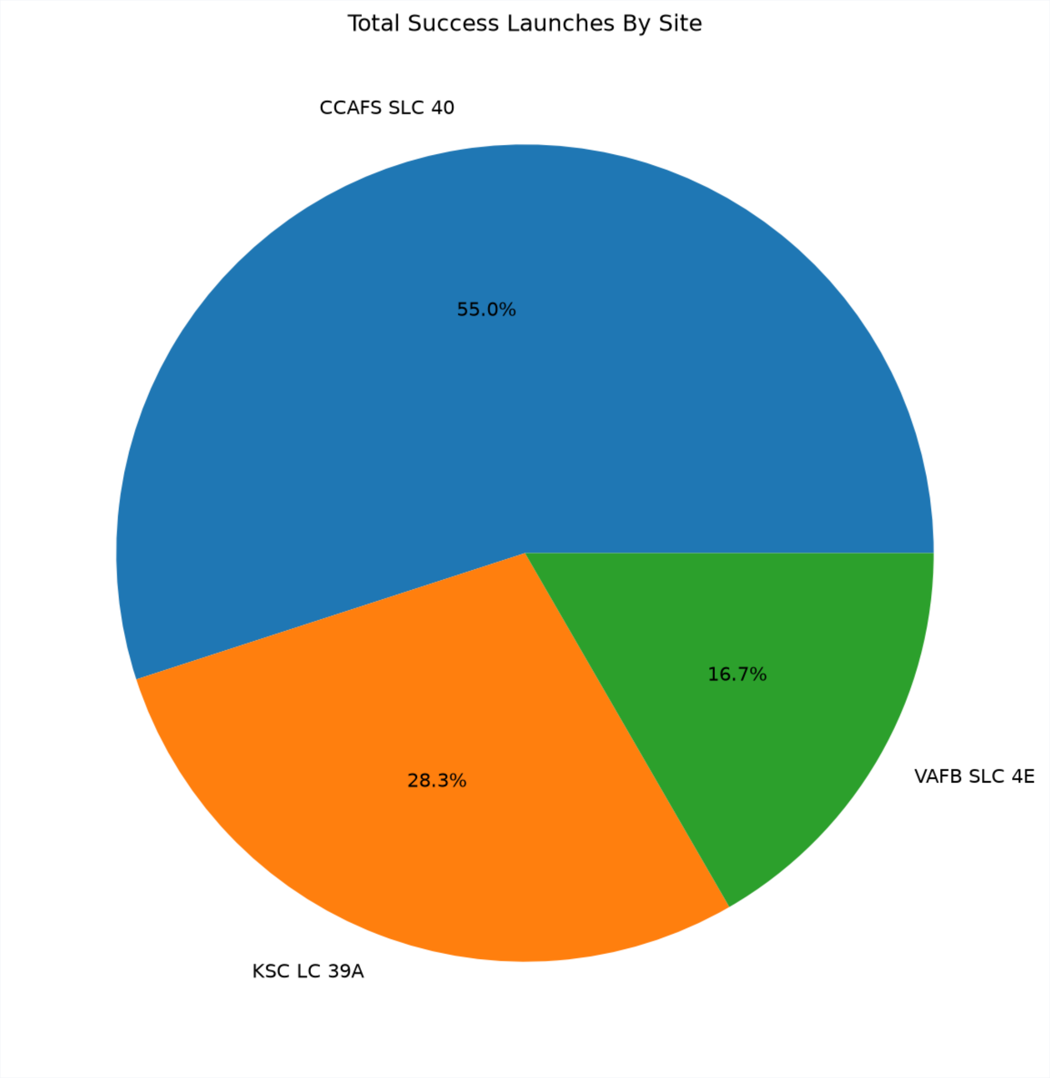
*Vikas Parmar*

Section 4

# Build a Dashboard with Plotly Dash

Vikas Parmar

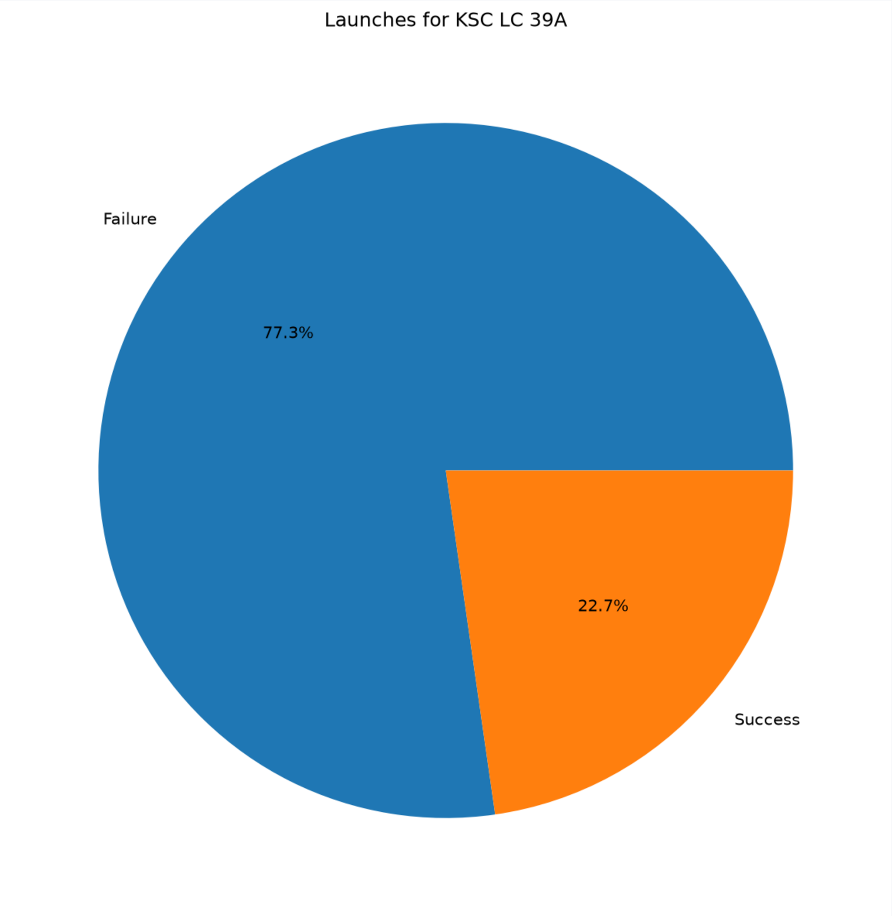
# Dashboard Screenshot 1




Success share by launch site across all locations.

*Vikas Parmar*

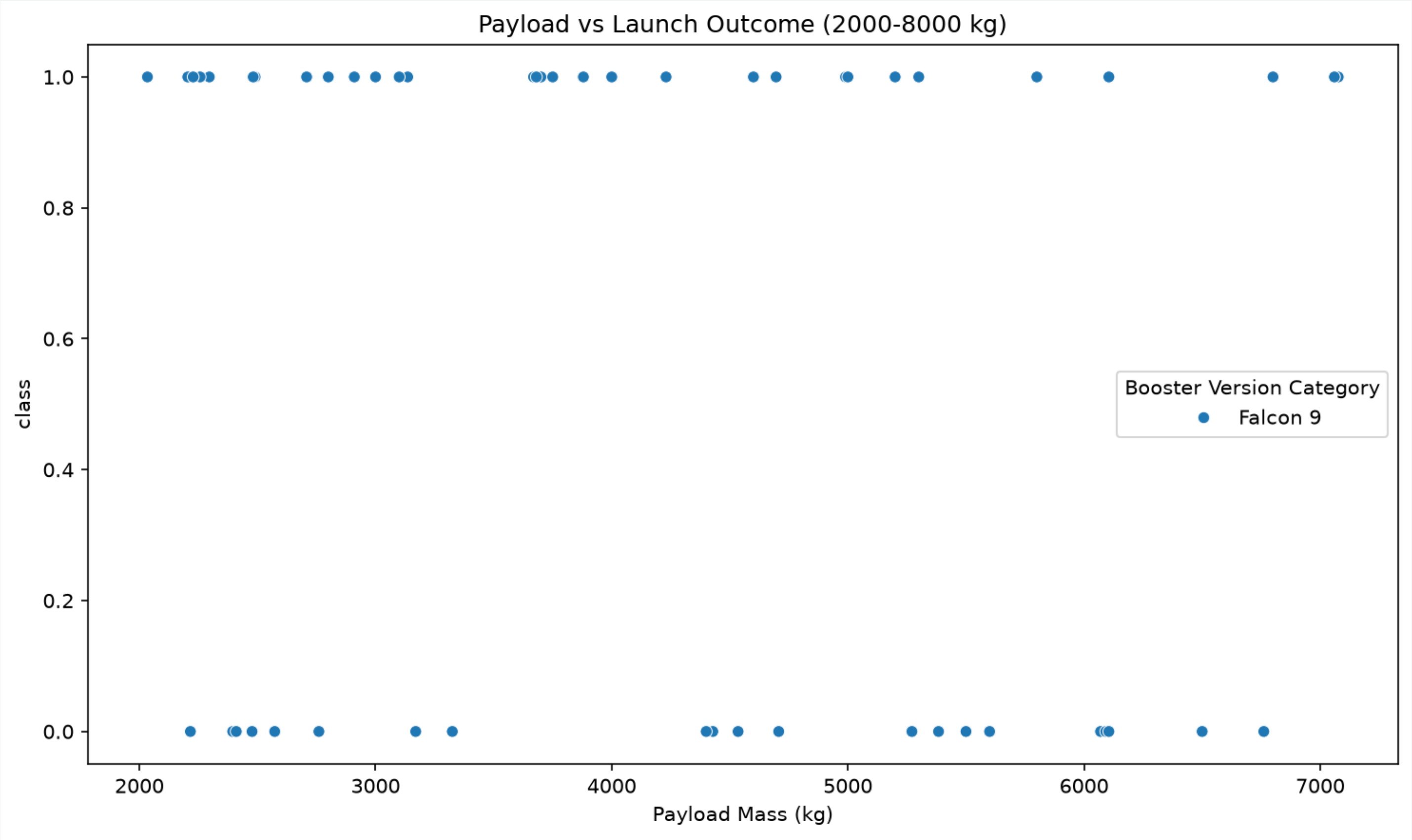
# Dashboard Screenshot 2




Site with highest success ratio shown in pie chart.

*Vikas Parmar*

# Dashboard Screenshot 3




Payload vs outcome for selected payload range.

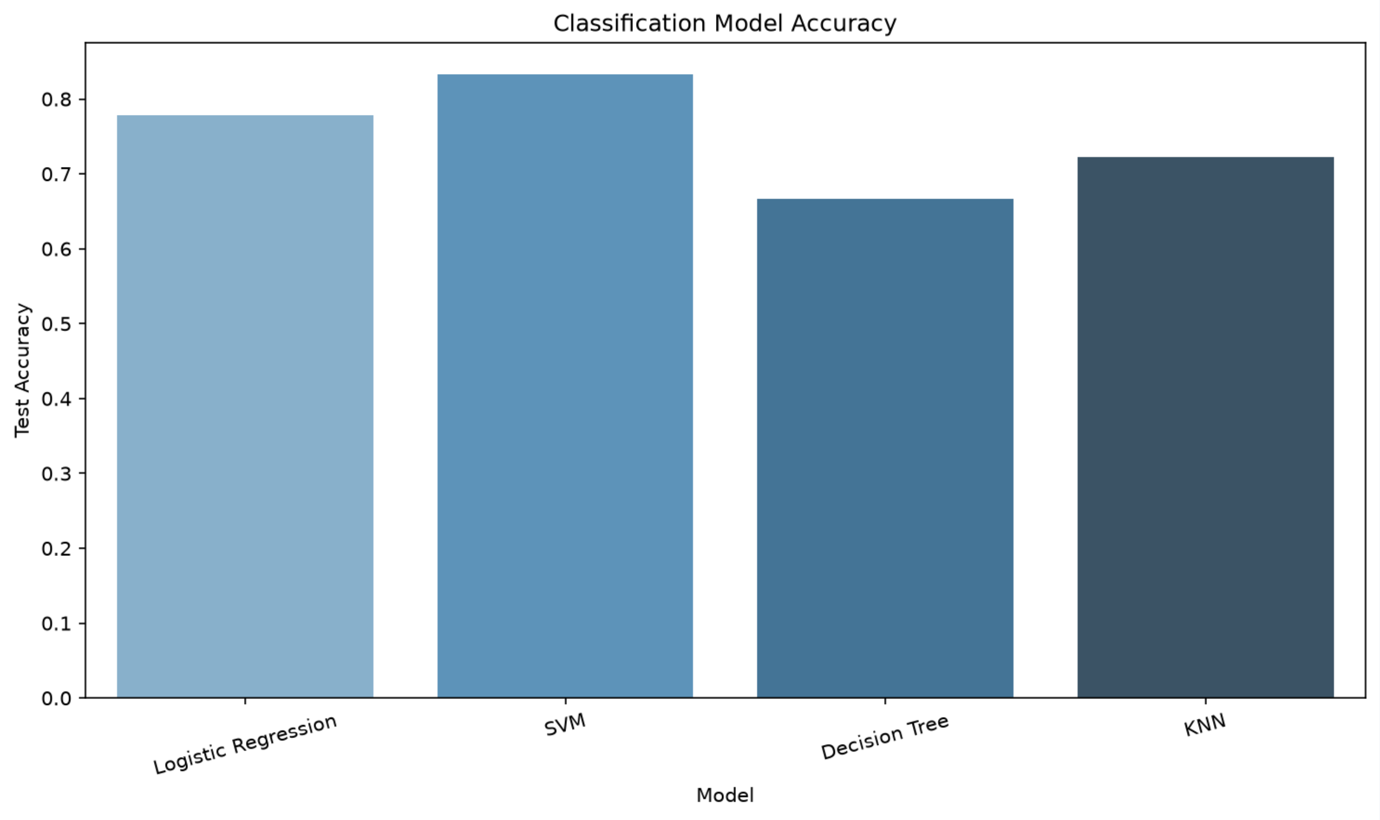
Vikas Parmar

Section 5

# Predictive Analysis (Classification)

Vikas Parmar

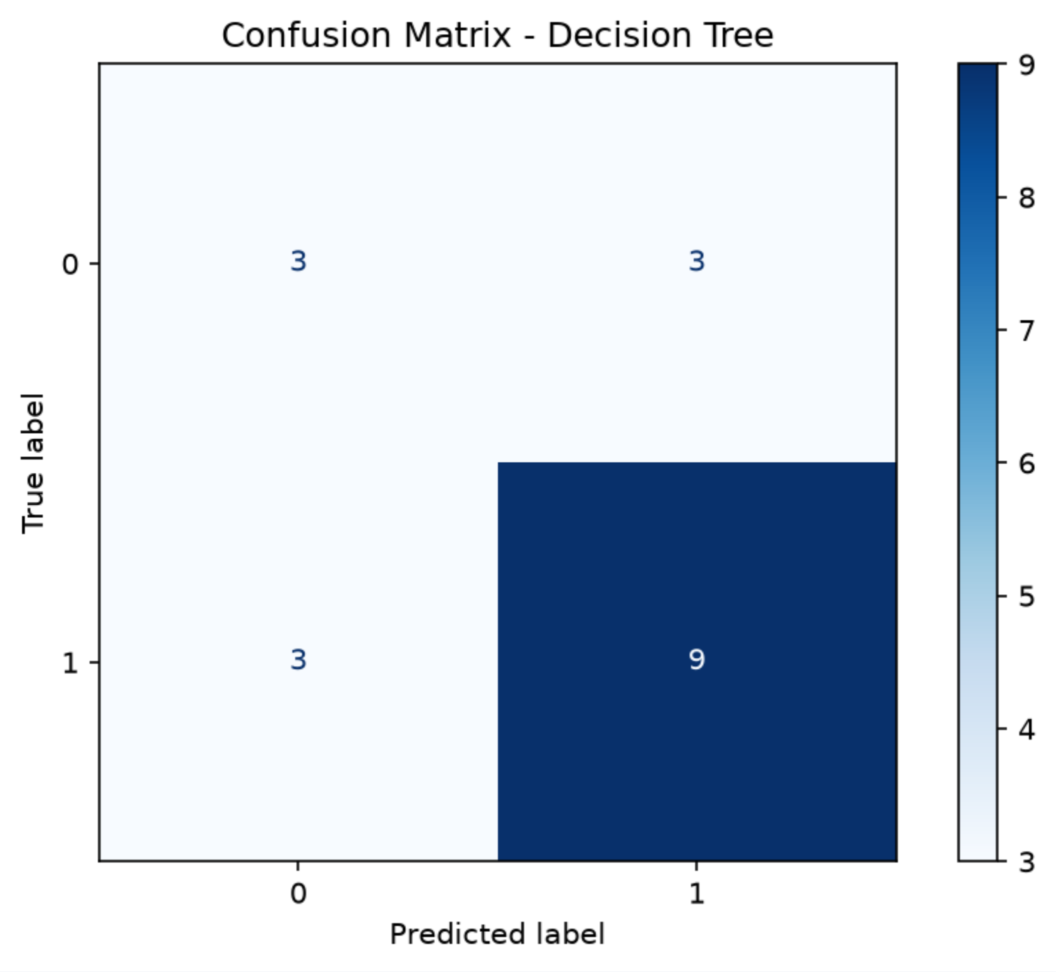
# Classification Accuracy




Model comparison shows test accuracy for all four classifiers. Decision Tree has the highest GridSearchCV score.

*Vikas Parmar*

# Confusion Matrix




Confusion matrix for Decision Tree. False positives would overestimate reusable first-stage availability and understate launch cost.

# Conclusions

1. Landing success improved significantly after 2014 as SpaceX gained experience.
2. Orbit type and payload mass strongly influence landing outcome.
3. Launch site selection affects both success rate and operational risk.
4. Decision Tree is the recommended model for predicting first-stage landing success.

# Appendix

- Author: Vikas Parmar
- GitHub Repository: https://github.com/vikas-parmar/applied-data-science-capstone-ibm
- Additional assets: notebooks, SQL query outputs, charts, and CSV datasets.

# Thank you!

Vikas Parmar